EE561 Advanced Static Power Conversion · Multilevel Converters module

Potential Final Questions

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Four questions for a closed-book exam with the instructor's cheat sheet, each designed for about 20 minutes and 25 points. Questions 1–2 cover Part I (topologies and their analysis), questions 3–4 cover Part II (modulation and balancing). Every question asks for a derivation, a numerical result and an engineering interpretation; full solutions and a marking guide follow on the later pages. All numbers differ from the lecture examples.

Question 1 · Device stress in a three-level NPC leg

PART I · ANALYSIS — **25 pts**

A 3-level NPC leg with V_{dc} = 800 V uses sine-triangle PWM at unity power factor: $r = m \sin\theta$, $i_a = \hat{I} \sin\theta$, with m = 0.8, $\hat{I}$ = 80 A, f_{sw} = 8 kHz. Device models: switch $v = 0.9\ \text{V} + 10\ \text{m}\Omega \cdot i$; diode $v = 1.0\ \text{V} + 8\ \text{m}\Omega \cdot i$; turn-on plus turn-off energy of a switch $E_{on} + E_{off} = 2.5 \times 10^{-7}\ \text{J/(V·A)} \times V \times I$; diode recovery energy $E_{rr} = 0.6 \times 10^{-7}\ \text{J/(V·A)} \times V \times I$.

a. For the positive half-cycle, state which devices conduct in states P and O, and identify the commutating pair. [5]

b. Derive the average and RMS currents of S1, S2 and the clamp diode D5 as functions of m and $\hat{I}$, and evaluate them. [10]

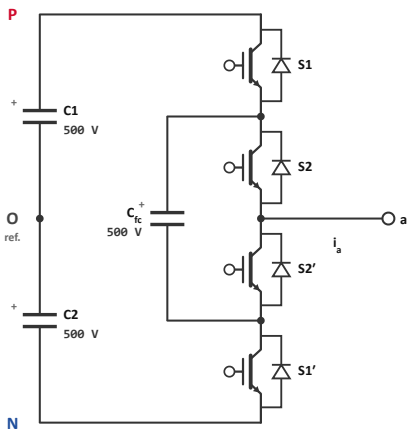
c. Compute the conduction and switching losses of S1, S2 and D5. Which device is the hottest? [6]

d. Repeat the ranking for m = 0.2 (a motor at low speed) and explain the change. How can an active NPC (ANPC) leg relieve the inner switches when the same converter works as a rectifier? [4]

Question 2 · Flying-capacitor leg: model and design

PART I · ANALYSIS AND DESIGN — **25 pts**

The 3-level flying-capacitor leg in the figure has V_{dc} = 1000 V and a flying-capacitor target V_{fc}^* = 500 V. Switches S1/S1′ and S2/S2′ are complementary pairs.



3-level flying-capacitor leg (O: reference midpoint only).

a. Tabulate the four states (S1 S2 = 11, 10, 01, 00) with the pole voltage v_{aO} in terms of V_{dc} and V_{fc}, and the capacitor current i_{fc} in terms of i_a. [6]

b. Write v_{aO} and i_{fc} as switching functions, average them over one switching period with duties d_1, d_2, and show that phase-shifted carriers ($d_1 = d_2$) give zero average capacitor current. [6]

c. With phase-shifted carriers a redundant state lasts at most half a switching period. Size C_{fc} for $\Delta V_{fc,pp} \le 10$ V at $\hat{I}$ = 60 A and f_{sw} = 20 kHz. [6]

d. V_{fc} is measured as 470 V while $i_a = -30$ A and the next output level is zero. Which state do you apply, and what pole voltage does it produce? [4]

e. Why must C_{fc} be pre-charged before PWM starts? [3]

Question 3 · Three-level space-vector modulation

PART II · MODULATION **25 pts**

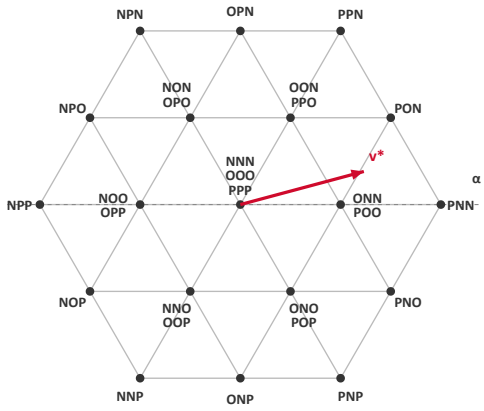
The figure shows the space-vector diagram of a 3-level NPC converter; vectors are in units of $V_{dc}/2$ (small vectors 2/3, medium 2/√3, large 4/3). The reference is **v*** = 0.85 ∠ 15°.

a. Identify the three nearest vectors (their switching states) and give their αβ coordinates. [4]

b. Using volt-second balance, compute the dwell-time fractions $d_k = T_k/T_s$. [8]

c. Construct a symmetric seven-segment sequence in which exactly one phase changes by one level at each transition. Give each segment's duration as a fraction of T_s (split the redundant small vector equally). [6]

d. Give the midpoint current i_O of every state in your sequence in terms of the phase currents. If $\Delta V = v_{C1} - v_{C2} > 0$ and $i_a > 0$, how should the small-vector time be redistributed (with C dΔV/dt = i_O, i_O drawn from O)? Does this change the output voltage? [7]



3-level space-vector diagram, states as phase levels (a b c).

Question 4 · Designing a neutral-point balancing loop

PART II · BALANCING AND CONTROL **25 pts**

A 3-level NPC with carrier PWM has a split DC link of 2 × 3.3 mF; an ideal source holds $v_{C1} + v_{C2}$ constant, so $C\,d\Delta V/dt = i_O$ with $\Delta V = v_{C1} - v_{C2}$ and i_O drawn from the midpoint. The phase currents are balanced with $\hat{I}$ = 40 A, and m = 0.9. A common offset r_0 is added to all references.

a. Starting from $\langle i_O \rangle = \Sigma_x(1 - |r_x + r_0|)\, i_x$, derive $\partial\langle i_O\rangle/\partial r_0$ and its average over a fundamental period for $r_x = m\cos\theta_x$, $i_x = \hat{I}\cos(\theta_x - \phi)$. [6]

b. With $r_0 = K \cdot \mathrm{sign}(P) \cdot \Delta V$, derive the closed-loop time constant and choose K for τ = 20 ms at cos ϕ = 0.8. [6]

c. With the same K, what is τ at cos ϕ = 0.2? What happens in rectifier mode if sign(P) is omitted? [5]

d. At unity power factor the midpoint current contains a third harmonic of amplitude $8m\hat{I}/(5\pi)$. Estimate the peak-to-peak ripple of ΔV. Can — and should — the balancing loop remove this ripple? [8]

Solutions and marking guide

Solution 1 · Device stress in a three-level NPC leg — 25 pts

(a) Positive half-cycle ($i_a > 0$): P → S1 + S2; O → D5 + S2. The commutating pair is **S1 ↔ D5** (short loop C1–S1–D5); S2 stays on and conducts the whole half-cycle. [5]

(b) With duties $d_P = m \sin\theta$ and $d_O = 1 - m \sin\theta$ over $0 < \theta < \pi$:

$$I_{avg,S1} = (1/2\pi)\int_0^{\pi} m \sin\theta \cdot \hat{I} \sin\theta \, d\theta = m\hat{I}/4, \quad I_{rms,S1}{}^2 = (1/2\pi)\int_0^{\pi} m \sin\theta \cdot \hat{I}^2 \sin^2\theta \, d\theta = 2m\hat{I}^2/3\pi$$

$$I_{avg,S2} = \hat{I}/\pi, \quad I_{rms,S2}{}^2 = \hat{I}^2/4; \quad I_{avg,D5} = \hat{I}(1/\pi - m/4), \quad I_{rms,D5}{}^2 = \hat{I}^2(1/4 - 2m/3\pi)$$

	I_{avg}	I_{rms}	conduction	switching	total
S1	16.0 A	33.0 A	25.3 W	20.4 W	**45.6 W**
S2	25.5 A	40.0 A	38.9 W	0	38.9 W
D5	9.46 A	22.7 A	13.6 W	4.9 W	18.5 W

(c) $P_{cond} = V_0 I_{avg} + r\, I_{rms}{}^2$. Switching occurs in every carrier period of the positive half-cycle at the commutation voltage $V_{dc}/2 = 400$ V: $P_{sw,S1} = f_{sw} \cdot 2.5\times10^{-7} \cdot 400 \cdot \hat{I}/\pi = 20.4$ W, $P_{rr,D5} = f_{sw} \cdot 0.6\times10^{-7} \cdot 400 \cdot \hat{I}/\pi = 4.9$ W. Hottest: **S1 (45.6 W)**. [6]

(d) At $m = 0.2$: S1 = 6.3 + 20.4 = 26.7 W, D5 = 32.1 + 4.9 = 37.0 W, S2 = 38.9 W (unchanged). The zero-state duty $1 - m \sin\theta$ grows, so the inner switch and the clamp diode now carry most of the current; D5 overtakes S1. In rectifier operation the commutation moves to S3 ↔ D1, so the inner switches both conduct and switch; in an ANPC leg the zero state can use the active clamps (O_U/O_L paths) so that S5/S6 take the commutation and the loss is redistributed. [4]

Marking (b): S1 derivation 4, S2 3, D5 3 (2 for the formula, 1 for the number).

Solution 2 · Flying-capacitor leg — 25 pts

(a) [6]

S1 S2	v_{aO}	nominal	i_{fc} (into + terminal)
11	$+V_{dc}/2$	+500 V	0
10	$V_{dc}/2 - V_{fc}$	0 V	$+i_a$
01	$-V_{dc}/2 + V_{fc}$	0 V	$-i_a$
00	$-V_{dc}/2$	−500 V	0

(b) $v_{aO} = -V_{dc}/2 + S_1(V_{dc} - V_{fc}) + S_2 V_{fc}$, $i_{fc} = (S_1 - S_2)\, i_a$. Averaging: $\langle v_{aO}\rangle = -V_{dc}/2 + d_1(V_{dc} - V_{fc}) + d_2 V_{fc}$, $\langle i_{fc}\rangle = (d_1 - d_2)\, i_a$. Phase-shifted carriers with one reference give $d_1 = d_2 \rightarrow \langle i_{fc}\rangle = 0$ and $\langle v_{aO}\rangle = (2d - 1)V_{dc}/2$. [6]

(c) $\Delta V_{fc,pp} \le \hat{I}(T_s/2)/C_{fc} \rightarrow C_{fc} \ge \hat{I}/(2 f_{sw} \Delta V) = 60/(2 \cdot 20\,000 \cdot 10) =$ **150 µF** (voltage rating ≥ 500 V plus margin). [6]

(d) V_{fc} is low → we need $i_{fc} = (S_1 - S_2) i_a > 0$. With $i_a < 0$ this requires $S_1 - S_2 = -1$: **state 01**, $i_{fc} = +30$ A. It produces $v_{aO} = -500 + 470 =$ **−30 V** instead of 0. [4]

(e) With $V_{fc} = 0$, state 11 places $V_{dc} - V_{fc} = 1000$ V across S1′, a device rated for about $V_{dc}/2$: it would fail. A pre-charge circuit or a controlled start-up sequence brings V_{fc} to 500 V first. [3]

Solution 3 · Three-level space-vector modulation **25 pts**

(a) $\mathbf{v}^* = 0.85\angle 15° = (0.821, 0.220)$. Nearest vectors: small **ONN/POO** (0.667, 0); large **PNN** (1.333, 0); medium **PON** (1.000, 0.577). [4]

(b) β: $0.577\, d_{PON} = 0.220 \rightarrow d_{PON} = 0.381$; $d_S + d_L = 0.619$. α: $0.667\, d_S + 1.333\, d_L + 0.381 = 0.821 \rightarrow d_L = 0.041$, $d_S = 0.578$. [8]

(c) ONN (0.145) → PNN (0.020) → PON (0.190) → POO (0.289) → PON (0.190) → PNN (0.020) → ONN (0.145). Transitions: phase a O→P, phase b N→O, phase c N→O, then back in reverse; each phase switches once up and once down per period. [6]

(d) i_O: ONN = $+i_a$; PNN = 0; PON = $+i_b$; POO = $i_b + i_c = -i_a$. To reduce ΔV we need $\langle i_O \rangle < 0$: **lengthen POO and shorten ONN**, keeping their sum at 0.578. Both states produce the same space vector, so the output (line-to-line) voltage does not change — only the midpoint charge does. [7]

Solution 4 · Neutral-point balancing loop **25 pts**

(a) For an offset small enough not to change the sign of any r_x: $\partial\langle i_O\rangle/\partial r_0 = -\Sigma_x \operatorname{sign}(r_x)\, i_x$. Since $\langle \operatorname{sign}(\cos\theta)\cos(\theta-\phi)\rangle = (2/\pi)\cos\phi$ for each phase, the period average is $\mathbf{-(6/\pi)\,\hat{I}\cos\phi}$. [6]

(b) $C\, d\Delta V/dt = -(6/\pi)\hat{I}\cos\phi \cdot K \operatorname{sign}(P)\,\Delta V = -(6/\pi)\hat{I}|\cos\phi|\,K\,\Delta V \rightarrow \boldsymbol{\tau = \pi C/(6\hat{I}K|\cos\phi|)}$. For $\tau = 20$ ms: $K = \pi \cdot 3.3\times10^{-3}/(6 \cdot 40 \cdot 0.8 \cdot 0.02) = \mathbf{2.7\times10^{-3}\ V^{-1}}$ (an offset of 0.027 per 10 V of imbalance). [6]

(c) $\tau \propto 1/|\cos\phi|$: $\tau = 20$ ms $\times\, 0.8/0.2 = \mathbf{80\ ms}$. Without sign(P), $\cos\phi < 0$ in rectifier mode makes the loop gain positive: ΔV grows exponentially until saturation and an overvoltage trip. [5]

(d) $\hat{I}_{O3} = 8 \cdot 0.9 \cdot 40/(5\pi) = 18.3$ A at 150 Hz. Integrating, $\Delta V_{pp} \approx 2\hat{I}_{O3}/(3\omega C) = 2 \cdot 18.3/(3 \cdot 314 \cdot 3.3\times10^{-3}) \approx \mathbf{11.8\ V}$ (a numerical integration gives 11.9 V). The loop bandwidth is about $1/(2\pi\tau) \approx 8$ Hz, far below 150 Hz, so it cannot remove the ripple — and it should not try: the ripple is inherent to carrier PWM with zero average, and chasing it would only inject large offsets. It is handled by sizing C (or by modified modulation). [8]